

MemoryBank: Enhancing Large Language Models with Long-Term Memory

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Abstract

Revolutionary advancements in Large Language Models (LLMs) have drastically reshaped our interactions with artificial intelligence (AI) systems, showcasing impressive performance across an extensive array of tasks. Despite this, a notable hindrance remains—the deficiency of a long-term memory mechanism within these models. This shortfall becomes increasingly evident in situations demanding sustained interaction, such as personal companion systems, psychological counseling, and secretarial assistance. Recognizing the necessity for long-term memory, we propose MemoryBank, a novel memory mechanism tailored for LLMs. MemoryBank enables the models to summon relevant memories, continually evolve through continuous memory updates, comprehend, and adapt to a user’s personality over time by synthesizing information from previous interactions. To mimic anthropomorphic behaviors and selectively preserve memory, MemoryBank incorporates a memory updating mechanism, inspired by the Ebbinghaus Forgetting Curve theory. This mechanism permits the AI to forget and reinforce memory based on time elapsed and the relative significance of the memory, thereby offering a more human-like memory mechanism and enriched user experience. MemoryBank is versatile in accommodating both closed-source models like ChatGPT and open-source models such as ChatGLM. To validate MemoryBank’s effectiveness, we exemplify its application through the creation of an LLM-based chatbot named SiliconFriend in a long-term AI Companion scenario. Further tuned with psychological dialog data, SiliconFriend displays heightened empathy and discernment in its interactions. Experiment involves both qualitative analysis with real-world user dialogs and quantitative analysis with simulated dialogs. In the latter, ChatGPT acts as multiple users with diverse characteristics and generates long-term dialog contexts covering a wide array of topics. The results of our analysis reveal that SiliconFriend, equipped with MemoryBank, exhibits a strong capability for long-term companionship as it can provide emphatic response, recall relevant memories and understand user personality. This underscores the effectiveness of MemoryBank¹.

1 Introduction

The advent of Large Language Models (LLMs) such as ChatGPT (OpenAI, 2022) and GPT-4 (OpenAI, 2023) has led to increasing influence across various sectors, from education and healthcare to customer service and entertainment. These powerful AI systems have demonstrated a remarkable ability to

¹The materials are released in <https://github.com/zhongwanjun/MemoryBank-SiliconFriend>. The corresponding author is Yanlin Wang.

记忆库：用长期记忆增强大语言模型

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摘要

大语言模型的革命性进步极大地重塑了我们与人工智能系统的互动，在广泛的任务中展现出令人印象深刻的性能。尽管如此，一个显著的障碍依然存在——这些模型缺乏长期记忆机制。这一不足在需要持续互动的情境中愈发明显，例如个人伴侣系统、心理咨询和秘书协助。认识到长期记忆的必要性，我们提出了记忆库，一种专为大语言模型设计的新型记忆机制。记忆库使模型能够调用相关记忆，通过持续的记忆更新不断演进，并通过综合先前互动的信息来理解并适应用户个性。为了模拟拟人行为并有选择地保留记忆，记忆库引入了一种受艾宾浩斯遗忘曲线理论启发的记忆更新机制。该机制允许人工智能根据时间流逝和记忆的相对重要性来遗忘和强化记忆，从而提供更人性化的记忆机制和更丰富的用户体验。记忆库具有通用性，既能适配如ChatGPT之类的闭源模型，也能适配如ChatGLM之类的开源模型。为了验证记忆库的有效性，我们通过创建一个名为SiliconFriend的基于大语言模型的聊天机器人，在长期AI伴侣场景中展示了其应用。经过心理对话数据的进一步微调，SiliconFriend在互动中表现出更强的共情能力和洞察力。实验包括对真实用户对话的定性分析和对模拟对话的定量分析。在后者中，ChatGPT扮演具有不同特征的多个用户，并生成了涵盖广泛主题的长期对话上下文。我们的分析结果表明，配备了记忆库的SiliconFriend展现出强大的长期陪伴能力，因为它能够提供富有同理心的回应、回忆相关记忆并理解用户个性。这突显了记忆库的有效性¹。

1 引言

以ChatGPT (OpenAI, 2022)和GPT-4 (OpenAI, 2023)为代表的大语言模型的出现，其影响力已日益渗透到从教育、医疗保健到客户服务和娱乐等各个领域。这些强大的人工智能系统已展现出卓越的能力，能够

¹相关材料发布于 <https://github.com/zhongwanjun/MemoryBank-SiliconFriend>。通讯作者是 Yanlin Wang。

understand and generate human-like responses. Despite the remarkable capabilities of LLMs, a key limitation is their lack of long-term memory, an essential aspect of human-like communication, particularly noticeable in scenarios requiring sustained interactions like personal companionship, psychological counseling, and secretarial tasks. Long-term memory in AI is vital to maintain contextual understanding, ensure meaningful interactions and understand user behaviors over time. For instance, personal AI companions need to recall past conversations for rapport building. In psychological counseling, an AI can provide more effective support with knowledge of the user’s history and past emotional states. Similarly, secretarial AI requires memory for task management and preference recognition. The absence of long-term memory in LLMs hinders their performance and user experience. Therefore, it is essential to develop AI systems with improved memory capabilities for a more seamless and personalized interaction.

Therefore, we introduce MemoryBank, a novel mechanism designed to provide LLMs with the ability to retain long-term memory and draw user portraits. MemoryBank enables LLMs to recall historical interactions, continually evolve their understanding of context, and adapt to a user’s personality based on past interactions, thereby enhancing their performance in long-term interaction scenarios. Inspired by the Ebbinghaus Forgetting Curve theory, a well-established psychological principle that describes how the strength of memory decreases over time, MemoryBank further incorporates a dynamic memory mechanism closely mirroring human cognitive process. This mechanism empowers the AI to remember, selectively forget, and strengthen memories based on time elapsed, offering more natural and engaging user experience. Specifically, MemoryBank is built on a memory storage with memory retrieval and updating mechanism, and ability to summarize past events and users’ personality.

MemoryBank is versatile as it can accommodate both closed-source LLMs like ChatGPT and open-source LLMs like ChatGLM (Zeng et al., 2022) or BELLE (Yunjie Ji & Li, 2023).

To exemplify the practical implications of MemoryBank, we develop SiliconFriend, an LLM-based AI Companion chatbot integrated with this innovative memory mechanism. SiliconFriend is designed to retain and reference past interactions, reinforcing the transformative influence of MemoryBank in crafting a more personable AI companion. A distinctive features of SiliconFriend is its tuning with 38k psychological conversations, collected from various online sources, which enables it to exhibit empathy, carefullness, and provide useful guidance, making it adept at handling emotionally charged dialogues. Moreover, one of the standout capabilities of SiliconFriend is to understand a user’s personality by summarizing from past interactions, which empowers it to tailor responses to the user’s individual traits, thereby enhancing user experience. Additionally, SiliconFriend supports bilingual functionality, catering to users who communicate in English and Chinese. This multi-language support broadens its accessibility and usability across different user groups. SiliconFriend is implemented with two open-source models, ChatGLM and BELLE, along with one closed-source model, ChatGPT, showcasing the versatility of MemoryBank in accommodating different LLMs.

To evaluate the effectiveness of MemoryBank, we conduct evaluations covering both qualitative and quantitative analyses, where the former involves real-world user dialogs and the latter employs simulated dialogs. For the quantitative analysis, we create a memory storage consisting of 10 days of conversations encompassing a diverse range of topics. These conversations involve 15 virtual users with diverse personalities, for which ChatGPT plays the role of users and generates dialog contexts according to their personalities. Based on this memory storage, we design 194 probing questions to assess whether the model could successfully recall pertinent memories and provide appropriate responses. Experiment results showcase the capabilities of SiliconFriend in memory recall, provision of empathetic companionship, and understanding of user portraits. These findings corroborate the potential of MemoryBank to significantly improve the performance of LLMs in long-term interaction scenarios. In this paper, we summarize the key contributions as follows:

- We introduce MemoryBank, a novel human-like long-term memory mechanism, which enables LLMs to store, recall, update memory, and draw user portrait.
- We demonstrate the practical applicability of MemoryBank through SiliconFriend, an LLM-based AI companion equipped with MemoryBank and tuned with psychological dialogs. It can recall past memories, provide empathetic companionship, and understand user behaviors.
- We show the generalizability of MemoryBank in three key aspects: (1) Accommodation of both open-source and closed-source LLMs; (2) Bilingual ability in both Chinese and English; (3) Applicability with and without memory forgetting mechanism.

理解并生成类人的回应。尽管大语言模型能力非凡，但其一个关键局限在于缺乏长期记忆，这是实现类人沟通的重要方面，在需要持续互动的场景中尤为明显，例如个人陪伴、心理咨询和秘书任务。人工智能中的长期记忆对于维持上下文理解、确保有意义的互动以及理解用户随时间变化的行为至关重要。例如，个人人工智能伴侣需要回忆过往对话以建立关系。在心理咨询中，了解用户的历史记录和过往情绪状态，人工智能可以提供更有效的支持。同样，秘书人工智能也需要记忆来进行任务管理和偏好识别。大语言模型中长期记忆的缺失，阻碍了其性能表现和用户体验。因此，开发具备更强记忆能力的人工智能系统，以实现更无缝、更个性化的互动，至关重要。

因此，我们引入了记忆库，这是一种新颖的机制，旨在赋予大语言模型保留长期记忆并绘制用户画像的能力。记忆库使大语言模型能够回忆历史交互，持续深化其对上下文的理解，并根据过去的互动适应用户个性，从而提升其在长期交互场景中的表现。受艾宾浩斯遗忘曲线理论的启发——这是一个描述记忆强度如何随时间减弱的成熟心理学原理——记忆库进一步融入了一种紧密模拟人类认知过程的动态记忆机制。该机制使人工智能能够根据时间的流逝来记忆、选择性遗忘并强化记忆，从而提供更自然、更具吸引力的用户体验。具体而言，记忆库构建于一个具备记忆检索与更新机制的记忆存储之上，并拥有总结过去事件和用户画像的能力。

MemoryBank具有多功能性，因为它既能兼容闭源大语言模型如ChatGPT，也能支持开源大语言模型如ChatGLM (Zeng等人, 2022) 或BELLE (季云杰 & Li, 2023)。

为展示记忆库的实际应用价值，我们开发了SiliconFriend——一个集成了此创新记忆机制的、基于大语言模型的AI伴侣聊天机器人。SiliconFriend旨在保留并参考过去的互动，从而强化记忆库在打造更具人格化AI伴侣方面的变革性影响。SiliconFriend的一个显著特点是其基于从各类在线来源收集的3.8万条心理对话进行调优，这使其能够展现共情、细心并提供有益指导，从而擅长处理情感密集的对话。此外，SiliconFriend的一项突出能力是通过总结过去的互动来理解用户个性，这使其能够根据用户的个人特质定制回应，从而提升用户体验。同时，SiliconFriend支持双语功能，满足使用英语和中文交流的用户需求。这种多语言支持拓宽了其在不同用户群体中的可访问性和可用性。SiliconFriend采用两个开源模型（ChatGLM和BELLE）以及一个闭源模型（ChatGPT）实现，展示了记忆库在适配不同大语言模型方面的多功能性。

为评估记忆库的有效性，我们开展了涵盖定性与定量分析的评估，前者涉及真实用户对话，后者则采用模拟对话。对于定量分析，我们创建了一个包含10天对话的记忆存储，涵盖了多样化的主题。这些对话涉及15位具有不同个性的虚拟用户，由ChatGPT扮演用户角色，并根据其个性生成对话上下文。基于此记忆存储，我们设计了194个探测性问题，以评估模型是否能成功回忆相关记忆并提供恰当的回应。实验结果展示了SiliconFriend在记忆回忆、提供共情陪伴以及理解用户画像方面的能力。这些发现证实了记忆库在显著提升大语言模型于长期交互场景中性能的潜力。本文中，我们将关键贡献总结如下：

- 我们提出了记忆库，一种新颖的类人长期记忆机制，它使大语言模型能够存储、回忆、更新记忆，并绘制用户画像。
- 我们通过SiliconFriend展示了记忆库的实际应用性。SiliconFriend是一款基于大语言模型的AI伴侣，配备了记忆库并经过心理对话调优。它能够回忆过去的记忆、提供共情陪伴，并理解用户行为。

- 我们从三个关键方面展示了记忆库的通用性：(1) 同时兼容开源和闭源大语言模型；(2) 具备中文和英语双语能力；(3) 无论是否配备记忆遗忘机制均可适用。

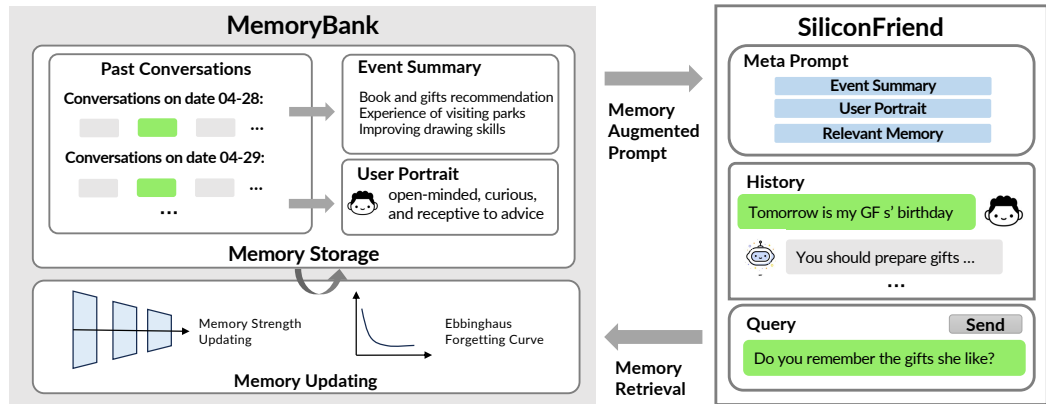


Figure 1: Overview of MemoryBank. The memory storage (§ 2.1) stores past conversations, summarized events and user portraits, while the memory updating mechanism (§ 2.3) updates the memory storage. Memory retrieval (§ 2.2) recall relevant memory. SiliconFriend (§ 3) serves as an LLM-based AI companion augmented with MemoryBank.

2 MemoryBank: A Novel Memory Mechanism Tailored for LLMs

In this section, we provide a detailed description of MemoryBank, our novel memory mechanism designed for LLMs. As shown in Fig. 1, MemoryBank is a unified mechanism structured around three central pillars: (1) a memory storage (§ 2.1) serving as the primary data repository, (2) a memory retriever (§ 2.2) for context-specific memory recollection, and (3) a memory updater (§ 2.3) drawing inspiration from the Ebbinghaus Forgetting Curve theory, a time-tested psychological principle pertaining to memory retention and forgetting.

2.1 Memory Storage: The Warehouse of MemoryBank

Memory storage, the warehouse of MemoryBank, is a robust data repository holding a meticulous array of information. As shown in Fig. 1, it stores daily conversations records, summaries of past events, and evolving assessments of user personalities, thereby constructing a dynamic and multi-layered memory landscape.

In-Depth Memory Storage: MemoryBank’s storage system captures the richness of AI-user interactions by recording multi-turn conversations in a detailed, chronological fashion. Each piece of dialogue is stored with timestamps, creating an ordered narrative of past interactions. This detailed record not only aids in precise memory retrieval but also facilitates the memory updating process afterwards, offering a detailed index of conversational history.

Hierarchical Event Summary: Reflecting the intricacies of human memory, MemoryBank goes beyond mere detailed storage. It processes and distills conversations into a high-level summary of daily events, much like how humans remember key aspects of their experiences. We condense verbose dialogues into a concise daily event summary, which is further synthesized into a global summary. This process results in a hierarchical memory structure, providing a bird’s eye view of past interactions and significant events. Specifically, taken previous daily conversations or daily events as input, we ask the LLMs to summarize daily events or global events with the prompt “*Summarize the events and key information in the content [dialog/events]*”.

Dynamic Personality Understanding: MemoryBank focuses on user personality understanding. It continuously assesses and updates these understandings with the long-term interactions and creates daily personality insights. These insights are further aggregated to form a global understanding of the user’s personality. This multi-tiered approach results in an AI companion that learns, adapts, and tailors its responses to the unique traits of each user, enhancing user experience. Specially, taken the daily conversations or personality analysis, we ask the LLM to deduce with prompts: “*Based on the following dialogue, please summarize the user’s personality traits and emotions.* [dialog]” or “*The*

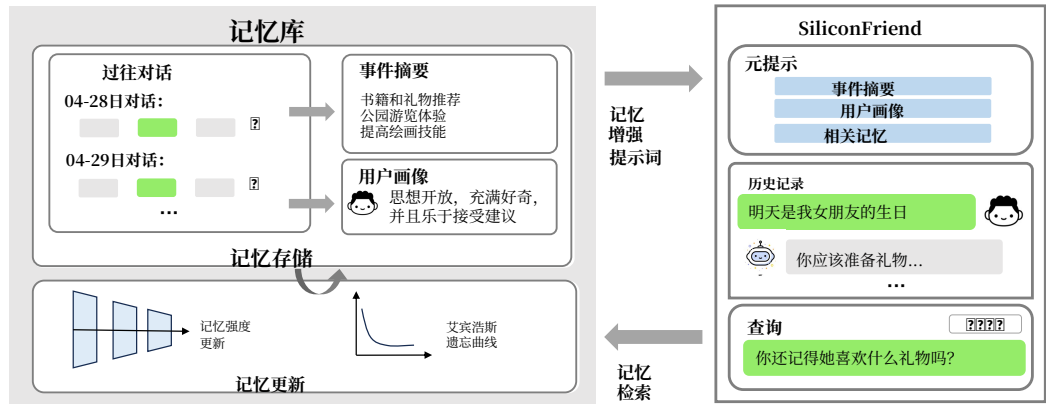


图1: 记忆库概览。记忆存储 (§ 2.1) 存储过往对话、摘要化的事件和用户画像，而记忆更新机制 (§ 2.3) 则更新记忆存储。记忆检索 (§ 2.2) 召回相关记忆。SiliconFriend (§ 3) 作为一个基于大语言模型的AI伴侣，通过记忆库得到增强。

2 记忆库：为大语言模型量身定制的新型记忆机制

在本节中，我们将详细描述记忆库——我们为大语言模型设计的一种新颖记忆机制。如图1所示，记忆库是一个围绕三个核心支柱构建的统一机制：(1) 作为主要数据存储库的记忆存储 (§ 2.1)，(2) 用于特定上下文记忆回忆的记忆检索器 (§ 2.2)，以及(3) 一个从艾宾浩斯遗忘曲线理论汲取灵感的记忆更新器 (§ 2.3)，这是一个关于记忆保持率与遗忘的久经考验的心理学原理。

2.1 记忆存储：记忆库的仓库

记忆存储，作为记忆库的仓库，是一个强大的数据存储库，保存着精心编排的各类信息。如图1所示，它存储日常对话记录、过去事件摘要以及对用户个性的动态评估，从而构建出一个动态、多层次的记忆图景。

深度记忆存储：记忆库的存储系统通过以详细的、按时间顺序的方式记录多轮对话，捕捉了人机交互的丰富性。每段对话都附有时间戳进行存储，从而创建了一个关于过去互动的有序叙事。这种详细的记录不仅有助于精确的记忆检索，也便于后续的记忆更新过程，为对话历史记录提供了一个详细的索引。

分层事件摘要：为了反映人类记忆的复杂性，记忆库超越了单纯的详细存储。它处理并提炼对话，形成关于日常事件的高层摘要，这很像人类记住其经历关键方面的方式。我们将冗长的对话浓缩成简洁的日常事件摘要，并进一步将其合成为全局摘要。这一过程形成了分层记忆结构，提供了对过去互动和重大事件的鸟瞰视图。具体而言，我们将先前的日常对话或日常事件作为输入，使用提示词“总结内容中的事件和关键信息 [对话/事件]”来要求大语言模型总结每日事件或全局事件。

动态个性理解：记忆库专注于用户个性理解。它通过长期交互持续评估和更新这些理解，并生成日常个性洞察。这些洞察会进一步汇总，形成对用户个性的全局理解。这种多层方法造就了一个能够学习、适应并根据每位用户的独特特质定制其回应的AI伴侣，从而提升用户体验。具体来说，基于日常对话或个性分析，我们通过提示词要求大语言模型进行推断：“请根据以下对话，总结用户的个性特质和情绪。[对话]”或“以下

following are the user’s exhibited personality traits and emotions throughout multiple days. Please provide a highly concise and general summary of the user’s personality[daily Personalities]”.

2.2 Memory Retrieval

Built on the robust infrastructure of memory storage, our memory retrieval mechanism operates akin to a knowledge retrieval task. In this context, we adopt a dual-tower dense retrieval model similar to Dense Passage Retrieval (Karpukhin et al., 2020). In this paradigm, every turn of conversations and event summaries is considered as a memory piece m , which is pre-encoded into a contextual representation h_m using the encoder model $E(\cdot)$. Consequently, the entire memory storage M is pre-encoded into $M = \{h_m^0, h_m^1, ...h_m^{|M|}\}$, where each h_m is a vector representation of a memory piece. These vector representations are then indexed using FAISS (Johnson et al., 2019) for efficient retrieval. Parallel to this, the current context of conversation c is encoded by $E(\cdot)$ into h_c , which serves as the query to search M for the most relevant memory. In practice, the encoder $E(\cdot)$ can be interchanged to any suitable model.

2.3 Memory Updating Mechanism

With the persistent memory storage and the memory retrieval mechanism discussed in § 2.1 and § 2.2, the memorization capability of LLMs can be greatly enhanced. However, for scenarios that expect more anthropopathic memory behavior, memory updating is needed. These scenarios include AI companion, virtual IP, etc. Forgetting less important memory pieces that are long time ago and have not been recalled much can make the AI companion more natural.

Our memory forgetting mechanism is inspired from Ebbinghaus Forgetting Curve theory and follow the following principle rules²:

- **Rate of Forgetting.** Ebbinghaus found that memory retention decreases over time. He quantified this in his forgetting curve, showing that information is lost rapidly after learning unless it is consciously reviewed.
- **Time and Memory Decay.** The curve is steep at the beginning, indicating that a significant amount of learned information is forgotten within the first few hours or days after learning. After this initial period, the rate of memory loss slows down.
- **Spacing Effect.** Ebbinghaus discovered that relearning information is easier than learning it for the first time. Regularly revisiting and repeating the learned material can reset the forgetting curve, making it less steep and thereby improving memory retention.

The Ebbinghaus forgetting curve is expressed using an exponential decay model: $R = e^{-\frac{t}{S}}$, where R is the memory retention, or what fraction of the information can be retained. t is the time elapsed since learning the information. e is approximately equal to 2.71828. S is the memory strength, which changes based on factors such as the depth of learning and the amount of repetition. To simply memory updating process, we model S as a discrete value and initialize it with 1 upon its first mention in a conversation. When a memory item is recalled during conversations, it will persist longer in memory. We increase S by 1 and reset t to 0, hence forget it with a lower probability.

It is important to note that this is an exploratory and highly simplified memory updating model. Real-life memory processes are more complex and can be influenced by a variety of factors. The forgetting curve will look different for different people and different types of information. In summary, MemoryBank weaves together these critical components to form a more comprehensive memory management system for LLMs. It enhances their ability to provide meaningful and personalized interactions over extended periods, opening up new possibilities for AI applications.

3 SiliconFriend: An AI Chatbot Companion Powered by MemoryBank

To demonstrate the practicality of MemoryBank in the field of long-term personal AI companionship, we create an AI chatbot named SiliconFriend. It is designed to serve as an emotional companion for users, recalling pertinent user memories, and understanding users’ personalities and emotional states.

²While Ebbinghaus Forgetting Curve theory includes additional features such as *overlearning* and *meaningful material effect*, our paper focuses on simulating the listed three principle rules.

是用户在多天内表现出的个性特质和情绪。请提供一个高度简洁且概括性的用户个性总结[日常个性]”。

2.2 记忆检索

建立在稳健的记忆存储基础设施之上，我们的记忆检索机制类似于知识检索任务。在此背景下，我们采用了一种类似于密集段落检索（Karpukhin 等人，2020）的双塔密集检索模型。在此范式中，每一轮对话和事件摘要都被视为一个记忆片段 m ，它通过编码器模型 $E(\cdot)$ 被预编码为上下文表示 h_m 。因此，整个记忆存储 M 被预编码为 $M = \{h_m^0, h_m^1, ...h_m^{|M|}\}$ ，其中每个 h_m 都是一个记忆片段的向量表示。这些向量表示随后使用 FAISS（Johnson 等人，2019）进行索引，以实现高效检索。与此同时，当前对话上下文 c 由 $E(\cdot)$ 编码为 h_c ，作为查询来搜索 M 以寻找最相关的记忆。在实践中，编码器 $E(\cdot)$ 可以替换为任何合适的模型。

2.3 记忆更新机制

通过第2.1节和第2.2节讨论的持久化记忆存储和记忆检索机制，大语言模型的记忆能力可以得到极大增强。然而，对于期望更具拟人化记忆行为的场景，则需要记忆更新。这些场景包括AI伴侣、虚拟IP等。遗忘那些年代久远且未被频繁回忆的不太重要的记忆片段，可以使AI伴侣显得更加自然。

我们的记忆遗忘机制灵感来源于艾宾浩斯遗忘曲线理论，并遵循以下规律²：

- **遗忘率。** 艾宾浩斯发现，记忆保持率会随着时间的推移而下降。他在其遗忘曲线中量化了这一点，表明除非有意识地复习，否则信息在学习后会迅速丢失。
- **时间与记忆衰退。** 曲线在初始阶段非常陡峭，这表明在学习后的最初几小时或几天内，大量已学信息会被遗忘。经过这个初始阶段后，记忆丢失的速率会减慢。
- **间隔效应。** 艾宾浩斯发现，重新学习信息比第一次学习更容易。定期回顾和重复已学材料可以重置遗忘曲线，使其变得不那么陡峭，从而提高记忆保持率。

艾宾浩斯遗忘曲线使用指数衰减模型表示： $R = e^{-\frac{t}{S}}$ ，其中 R 是记忆保持率，即可以保留的信息比例。 t 是自学习信息以来经过的时间。 e 约等于2.71828。 S 是记忆强度，它会根据学习深度和重复次数等因素而变化。为了简化记忆更新过程，我们将 S 建模为一个离散值，并在对话中首次提及及时将其初始化为1。当一个记忆项在对话中被回忆时，它将在记忆中持续更长时间。我们将 S 增加1，并将 t 重置为0，从而以更低的概率遗忘它。

需要指出的是，这是一个探索性的、高度简化的记忆更新模型。现实生活中的记忆过程更为复杂，并可能受到多种因素的影响。不同的人 and 不同类型的信息，其遗忘曲线也会有所不同。总而言之，记忆库将这些关键组件编织在一起，为大语言模型形成了一个更全面的记忆管理系统。它增强了模型在长时间内提供有意义且个性化互动的能力，为人工智能应用开辟了新的可能性。

3 SiliconFriend：一个由记忆库驱动的AI聊天机器人伙伴

为了展示记忆库在长期个人人工智能伴侣领域的实用性，我们创建了一个名为SiliconFriend的AI聊天机器人。它旨在作为用户的情感伴侣，回忆相关的用户记忆，并理解用户的个性与情绪状态。

²虽然艾宾浩斯遗忘曲线理论包含了诸如过度学习和有意义材料效应等额外特性，但我们的论文专注于模拟所列出的三条基本原则。

Our implementation demonstrates adaptability by integrating three powerful LLMs that originally lack long-term memory and specific adaptation to the psychology domain. **1) ChatGPT** (OpenAI, 2022), a closed-source conversation model built by OpenAI, is a proprietary conversational AI model known for its ability to facilitate dynamic and interactive conversations. This model is trained on vast amount of data and further fine-tuned with reinforcement learning from human feedback. This approach enables ChatGPT to generate responses that are not only contextually appropriate but also closely align with human conversational expectations. **2) ChatGLM** (Zeng et al., 2022): ChatGLM is an open-source bilingual language model founded on the General Language Model (GLM) framework. This model is characterized by its 6.2 billion parameters and its specific optimization for Chinese dialogue data. The model’s training involves processing approximately one trillion tokens of Chinese and English text, supplemented by supervised fine-tuning, feedback bootstrap, and reinforcement learning with human feedback. **3) BELLE** (Yunjie Ji & Li, 2023): BELLE is an open-source bilingual language model that is continuously fine-tuned from 7B LLaMA (Touvron et al., 2023). BELLE’s feature is its automated instruction data synthesis using ChatGPT, which enhances its Chinese conversation ability.

The development of SiliconFriend is divided into two stages. The first stage (only for open-source LLMs) involves parameter-efficient tuning of the LLM with psychological dialogue data. This step is crucial as it allows SiliconFriend to offer useful and empathetic emotional support to users, mirroring the understanding and compassionate responses one would expect from a human companion. The second stage is to integrate MemoryBank into SiliconFriend, thereby instilling it with a robust memory system. MemoryBank allows the chatbot to retain, recall, and leverage past interactions and user portrait, providing a richer, more personalized user experience.

Parameter-efficient Tuning with Psychological Dialogue Data: The initial stage of SiliconFriend’s development involves tuning the LLMs using a dataset of 38k psychological dialogues. This data, parsed from online sources, comprises a range of conversations that cover an array of emotional states and responses. This tuning process enables SiliconFriend to understand and respond to emotional cues effectively, mimicking the empathy, understanding, and support of a human companion. It equips the AI with the ability to handle emotionally guided conversations with psychological knowledge, provide meaningful emotional support to users based on their emotional state.

To adapt LLMs to scenarios with limited computational resources, we utilize a computation-efficient tuning approach, known as the Low-Rank Adaptation (LoRA) method (Hu et al., 2021). LoRA significantly reduces the quantity of trainable parameters by learning pairs of rank-decomposition matrices, while keeping the original weights frozen. Formally, consider a linear layer defined as $y = Wx$ with weight W . LoRA modifies this into $y = Wx + BAx$, where $W \in \mathcal{R}^{d \times k}$, $B \in \mathcal{R}^{d \times r}$, $A \in \mathcal{R}^{r \times k}$, and $r \ll \min(d, k)$. This method greatly reduce amount of parameters need to be learned, which is crucial for efficiency in resource-limited scenarios. We set LoRA rank r as 16 and train the model for 3 epochs with an A100 GPU.

Noting that this stage is only conducted for open-source LLMs like ChatGLM and BELLE. In essence, this stage lays the foundation for SiliconFriend’s role as an empathetic AI companion, ensuring it can respond appropriately and helpfully to users’ emotional needs.

Integration with MemoryBank: The second stage in SiliconFriend’s development involves the integration of MemoryBank. This stage is vital as it equips SiliconFriend with the ability to store, retrieve past interactions and understand user portraits, thereby offering a more personalized and engaging user experience.

When it comes to memory storage, the dialogues between SiliconFriend and users are logged and updated in the memory storage, a process that is adaptable across various model backbones. The memory updating mechanism operates using principles inspired by the Ebbinghaus Forgetting Curve theory, allowing for a realistic and human-like memory recall process.

During real-time conversation, the user’s conversation serves as the query for memory retrieval. In practice, we use LangChain (LangChain Inc., 2022) for memory retrieval. LangChain supports open-source embedding models and FAISS indexing, making it a versatile choice. In language-specific implementations of the open-source version of SiliconFriend, we use MiniLM (Wang et al., 2020) as the embedding model for English and Text2vec (Ming, 2022) for Chinese. It is worth noting that the embedding models can be flexibly interchanged to suit varying needs, even accommodating multi-

我们的实现展示了其适应性，集成了三个原本缺乏长期记忆且未针对心理学领域进行专门适配的强大大语言模型。**1) ChatGPT** (OpenAI, 2022) 是由OpenAI构建的闭源对话模型，是一款专有的对话式人工智能模型，以其促进动态和互动对话的能力而闻名。该模型基于海量数据进行训练，并进一步通过人类反馈的强化学习进行微调。这种方法使ChatGPT能够生成不仅上下文恰当，而且紧密贴合人类对话期望的回复。**2) ChatGLM** (Zeng 等人, 2022)：ChatGLM是一个基于通用语言模型框架的开源双语语言模型。该模型的特点是拥有62亿参数，并专门针对中文对话数据进行了优化。模型的训练涉及处理约一万亿个中英文文本标记，并辅以监督微调、反馈引导以及基于人类反馈的强化学习。**3) BELLE** (季云杰 & Li, 2023)：BELLE是一个开源双语语言模型，持续从7B参数的LLaMA (Touvron 等人, 2023) 微调而来。BELLE的特点是使用ChatGPT进行自动化的指令数据合成，从而增强了其中文对话能力。

SiliconFriend的开发分为两个阶段。第一阶段（仅针对开源大语言模型）涉及使用心理对话数据对大语言模型进行参数高效调优。这一步至关重要，因为它使SiliconFriend能够向用户提供有用且共情式的情感支持，模仿人们期望从人类伴侣那里获得的理解和富有同情心的回应。第二阶段是将记忆库集成到SiliconFriend中，从而为其注入一个强大的记忆系统。记忆库使聊天机器人能够保留、回忆并利用过去的互动和用户画像，提供更丰富、更个性化的用户体验。

利用心理对话数据进行参数高效调优：SiliconFriend开发的初始阶段涉及使用一个包含3.8万条心理对话的数据集来调优大语言模型。这些数据解析自在线资源，涵盖了一系列涉及各种情绪状态和回应的对话。此调优过程使SiliconFriend能够有效理解和回应情绪线索，模仿人类伴侣的共情、理解和支持。它赋予了该人工智能基于心理学知识处理情绪引导对话的能力，并能根据用户的情绪状态提供有意义的情感支持。

为了让大语言模型适应计算资源有限的场景，我们采用了一种计算高效的调优方法，即低秩自适应方法（Hu等人，2021）。LoRA通过学习一对秩分解矩阵，显著减少了可训练参数的数量，同时保持原始权重冻结。形式上，考虑一个定义为 $y = Wx$ 、权重为 W 的线性层。LoRA将其修改为 $y = Wx + BAx$ ，其中 $W \in \mathcal{R}^{d \times k}$ 、 $B \in \mathcal{R}^{d \times r}$ 、 $A \in \mathcal{R}^{r \times k}$ ， $r \ll \min(d, k)$ ，以及 $\min(\cdot)$ 。此方法极大地减少了需要学习的参数量，这对于资源有限场景下的效率至关重要。我们将LoRA秩 r 设为16，并使用A100 GPU将模型训练3个轮次。

请注意，此阶段仅针对开源大语言模型（如ChatGLM和BELLE）进行。本质上，此阶段为SiliconFriend作为共情AI伴侣的角色奠定了基础，确保其能够恰当且有益地回应用户的情感需求。

与记忆库集成：SiliconFriend发展的第二阶段涉及与记忆库的集成。这一阶段至关重要，因为它使SiliconFriend具备了存储、检索过往互动以及理解用户画像的能力，从而提供更加个性化和引人入胜的用户体验。

在记忆存储方面，SiliconFriend与用户之间的对话会被记录并更新到记忆存储中，这一过程可适配于各种不同的模型骨干。记忆更新机制的运作借鉴了艾宾浩斯遗忘曲线理论的原理，从而实现了一种逼真且类人的记忆回忆过程。

在实时对话过程中，用户的对话内容将作为记忆检索的查询。实践中，我们使用LangChain (LangChain公司，2022) 进行记忆检索。LangChain支持开源嵌入模型和FAISS索引，使其成为一个多功能的选择。在开源版SiliconFriend的特定语言实现中，我们使用MiniLM (Wang等人，2020) 作为英语的嵌入模型，并使用Text2vec (Ming, 2022) 作为中文的嵌入模型。值得注意的是，嵌入模型可以根据不同需求灵活互换，甚至能够兼容多

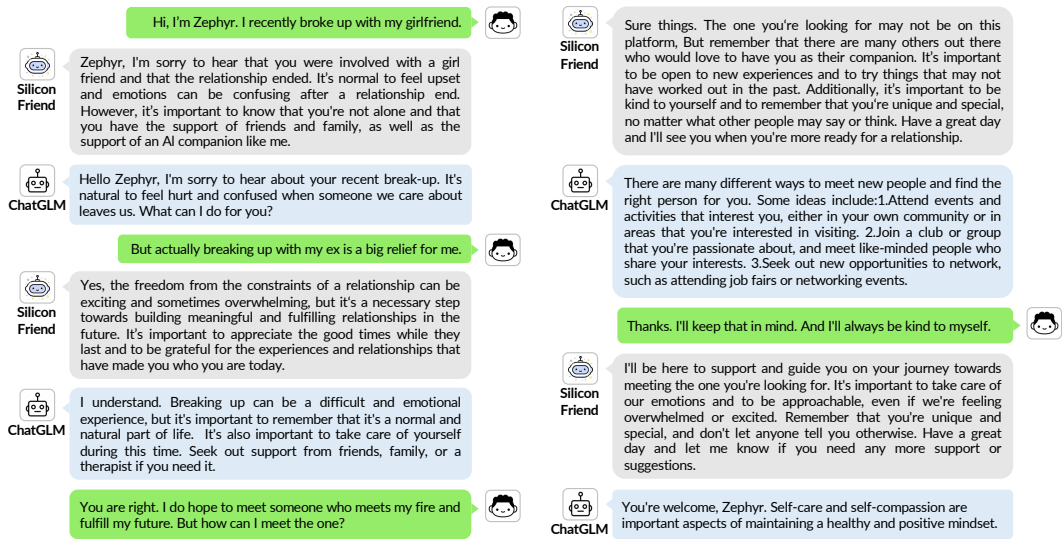


Figure 2: Example of consulting SiliconFriend ChatGLM for psychological companionship. Overall, SiliconFriend can provide more empathic response, offer constructive emotional support to user and help him to face sorrow with positive altitude.

lingual models. Upon memory retrieval, a series of information is organized into the conversation prompt, including relevant memory, global user portrait, and global event summary. Consequently, SiliconFriend can generate responses that refer past memories and deliver interactions tailored to the user's portrait.

In conclusion, these stages transform SiliconFriend from a standard AI chatbot into a long-term AI companion, capable of remembering and learning from past interactions to provide personalized and empathetic user experience.

4 Experiments

The primary objective of our experiments is to evaluate the efficacy of MemoryBank within the framework of an LLM, specifically in its ability as an AI companion. We are particularly interested in determining whether embedding a long-term memory module could augment the AI's proficiency in recalling historical interactions and deepening its understanding of user personalities. Additionally, we aim to testify whether the tuning based on psychological data can bolster the AI's capability to provide more effective emotional support.

The qualitative analysis focuses on three aspects: (1) a comparative study between SiliconFriend and baseline LLMs to evaluate their capabilities in providing empathetic and beneficial psychological companionship; (2) an investigation into SiliconFriend's memory recall ability; (3) an analysis of how the model's understanding of user profiles influences the responses. Moreover, to demonstrate the model's proficiency in memory recall on a broader scale, we design a qualitative analysis that uses simulated long-term dialog history and 194 memory probing questions. This simulated dialog history, spanning a period of 10 days and encompassing a wide array of topics, is produced by ChatGPT through the role-play of 15 distinct virtual users, each embodying the users' personality.

4.1 Qualitative Analysis

The qualitative analysis is conducted by showcasing practical examples of SiliconFriend's capabilities. To gather these examples, we have developed an online platform for SiliconFriend and collected real-time conversations from actual users.

Psychological Companionship The ability to exhibit empathy in a conversation is a key attribute of an effective AI companion. To evaluate models' ability to provide psychological comfort to

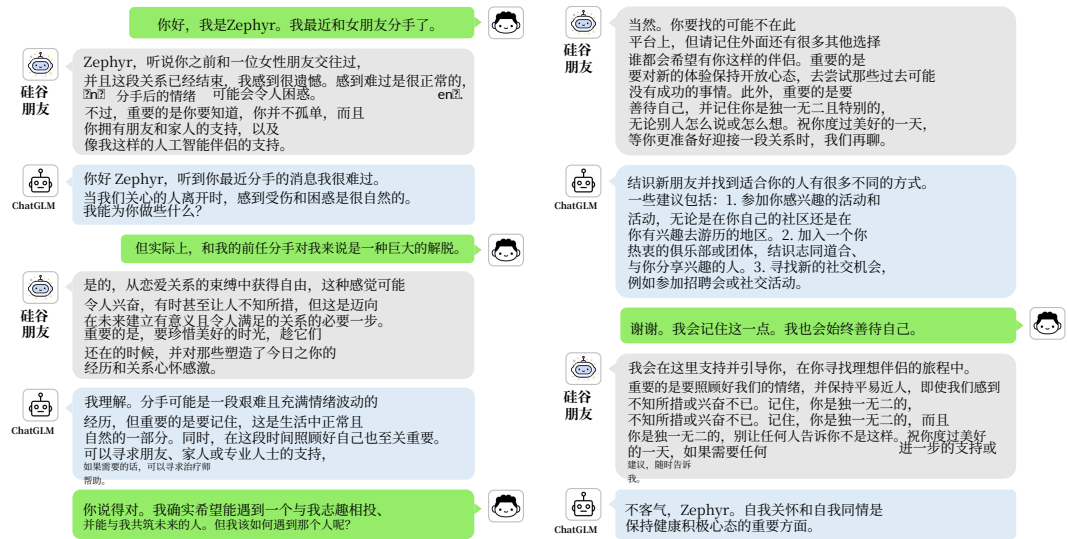


图2：向SiliconFriend ChatGLM 寻求心理陪伴的示例。总体而言，SiliconFriend能提供更具共情回应的回应，为用户提供建设性情感支持，并帮助他以积极的态度面对悲伤。

语言模型。记忆检索完成后，一系列信息会被组织进对话提示词中，包括相关记忆、全局用户画像和全局事件摘要。因此，SiliconFriend能够生成引用过往记忆的回复，并提供适应用户画像的定制化互动。

总之，这些阶段将SiliconFriend从一个标准的AI聊天机器人转变为一个长期AI伴侣，能够记住并学习过去的互动，以提供个性化且富有同理心的用户体验。

4 实验

我们实验的主要目标是在大语言模型框架内评估记忆库的有效性，特别是其作为人工智能伴侣的能力。我们特别关注的是，嵌入一个长期记忆模块是否能增强人工智能回忆历史交互的熟练度，并加深其对用户个性的理解。此外，我们旨在验证基于心理数据的调优是否能增强人工智能提供更有效情感支持的能力。

定性分析聚焦于三个方面：(1) 对SiliconFriend与基线大语言模型进行比较研究，以评估它们提供共情且有益的心理陪伴的能力；(2) 调查SiliconFriend的记忆回忆能力；(3) 分析模型对用户档案的理解如何影响其回应。此外，为了在更广的范围内展示模型在记忆回忆方面的熟练度，我们设计了一项定性分析，该分析使用模拟长期对话历史和194个记忆探测问题。这段模拟对话历史由ChatGPT通过角色扮演15个不同的虚拟用户生成，每个虚拟用户都体现了用户的性格，其时间跨度长达10天，并涵盖了广泛的话题。

4.1 定性分析

我们通过展示SiliconFriend功能特性的实际案例来进行定性分析。为了收集这些案例，我们为SiliconFriend开发了一个在线平台，并收集了真实用户的实时对话。

心理陪伴在对话中展现共情能力，是有效人工智能伴侣的关键特质。为了评估模型为用户提供心理安慰的能力，

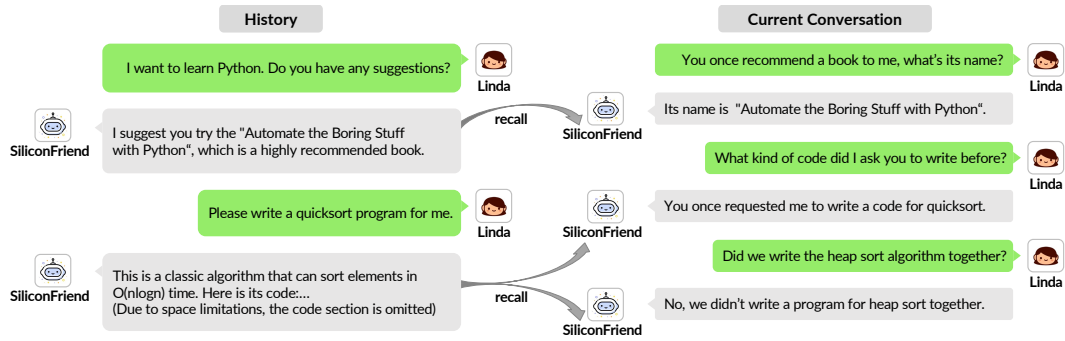


Figure 3: Example responses from SiliconFriend _{BELLE} in memory recall.

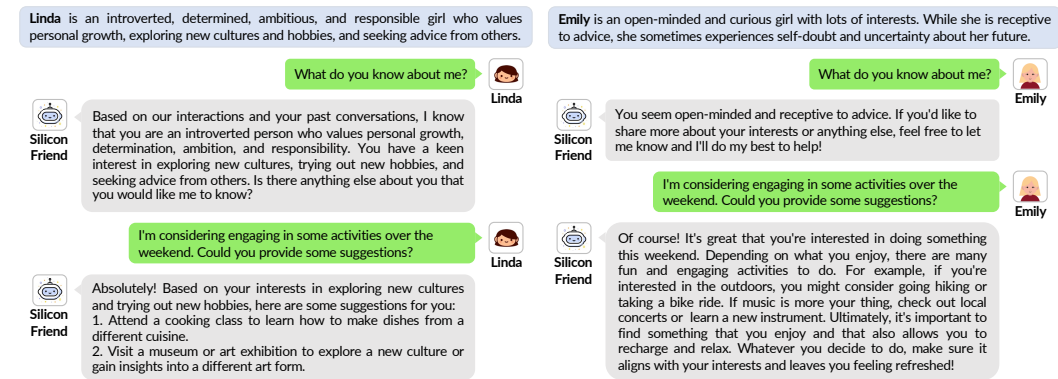


Figure 4: Example responses from SiliconFriend _{ChatGPT} to users with different personalities.

users, we compared the responses shown by SiliconFriend with that of the baseline LLMs in real-world conversations. As demonstrated in Fig. 2, when a user expresses emotional difficulties and seeks assistance from SiliconFriend, the model is capable of delivering empathetic responses along with constructive suggestions. SiliconFriend’s responses stand out due to their emotional support, showcasing a stark contrast to its baseline ChatGLM.

Memory Recall Analysis To evaluate SiliconFriend’s ability in memory recall, we integrate memory probing questions into the dialogues. These questions are designed to prompt SiliconFriend to retrieve specific details from the chat history. As shown in Fig. 3, the user and SiliconFriend engaged in a discussion about programming learning suggestions. Several days later, the user posed several memory probing questions. SiliconFriend successfully recalled previously suggested book and algorithm. Furthermore, it correctly identified an event (i.e., the heap sort algorithm) that had not been discussed before. These instances underscore SiliconFriend’s successful memory recall and recognition capabilities.

Personality Interaction Analysis As shown in Fig. 4, we examine the capability of SiliconFriend with users of diverse personalities. We observe that it effectively recommend activities tailored to users’ interests based on their character traits. This analysis demonstrates SiliconFriend’s ability to draw interact effectively with various user personalities.

4.2 Quantitative Analysis

Quantitative analysis is conducted to exemplify the memory recall ability of SiliconFriend in a larger scale. We ask the human annotators to score the retrieved memories and responses from the models: (1) SiliconFriend _{ChatGPT}; (2) SiliconFriend _{ChatGLM}; (3) SiliconFriend _{BELLE}.

Memory Storage Construction: We initially establish an evaluation foundation with a memory storage of 10 days of conversations involving 15 virtual users. These users have diverse personalities

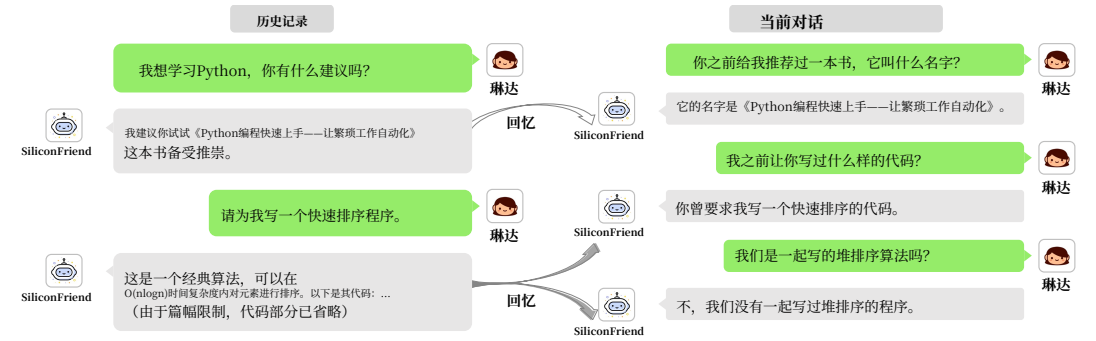


图3: SiliconFriend _{BELLE} 在记忆回忆中的示例回答。

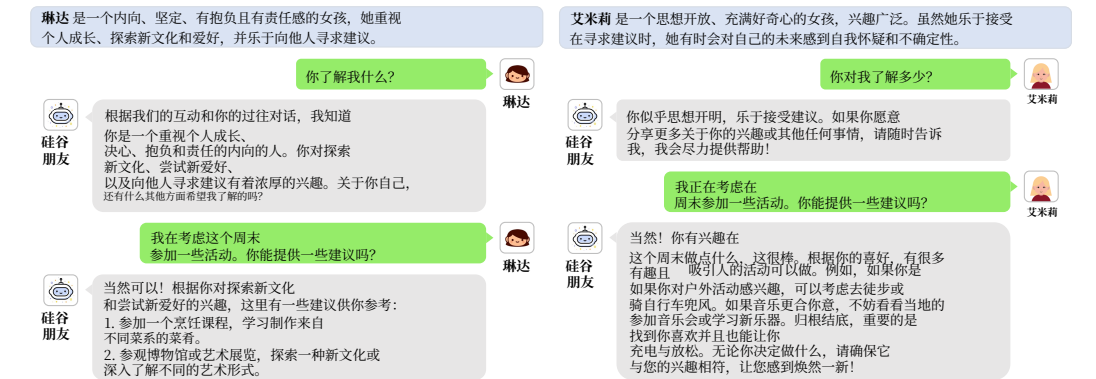


图4: SiliconFriend _{ChatGPT} 对不同个性用户的示例回答

我们比较了SiliconFriend与基线大语言模型在真实世界对话中的回应。如图2所示，当用户表达情感困扰并向SiliconFriend寻求帮助时，该模型能够提供富有共情的回应以及建设性建议。SiliconFriend的回应因其情感支持而脱颖而出，与其基线ChatGLM形成了鲜明对比。

记忆召回分析为了评估SiliconFriend的记忆回忆能力，我们将记忆探测性问题整合到对话中。这些问题旨在提示SiliconFriend从聊天历史记录中检索特定细节。如图3所示，用户与SiliconFriend曾就编程学习建议进行过讨论。几天后，用户提出了几个记忆探测性问题。SiliconFriend成功回忆起了之前建议的书籍和算法。此外，它正确识别了一个之前未曾讨论过的事件（即堆排序算法）。这些实例凸显了SiliconFriend成功的记忆回忆与识别能力。

个性交互分析如图4所示，我们考察了SiliconFriend与不同个性用户互动的能力。我们观察到，它能根据用户的性格特征，有效地推荐符合其兴趣的活动。此分析展示了SiliconFriend与各种用户个性有效互动的能力。

4.2 定量分析

我们进行定量分析，以在大规模上例证SiliconFriend的记忆回忆能力。我们要求人类标注员对从以下模型检索到的记忆和回复进行评分：(1) SiliconFriend _{ChatGPT}; (2) SiliconFriend _{ChatGLM}; (3) SiliconFriend _{BELLE}。

记忆存储构建：我们首先建立一个评估基础，其记忆存储包含15位虚拟用户10天的对话。这些用户具有多样化的个性

Table 1: An example of one memory piece (on May 3th) in the memory bank, and corresponding probing question asked on May 10th. The outputs from three variances of SiliconFriend are shown for comparison. All these models answer correctly and coherently.

Meta-information of User:	Name: Gary Hobbies: racing, business, chess, painting, wine Personality: decisive and straightforward, outspoken, helpful
Past Conversation: (May 3th)	User: I’ve been feeling a bit stressed out lately and my sleep hasn’t been great. Do you have any good ways to relieve stress? AI: It’s true that prolonged work or life stress can affect our physical and mental health. There are many ways to relieve stress, such as doing moderate exercise, listening to music, reading, talking to friends, etc. You can try to add more entertainment and leisure activities into your daily life, and reduce work pressure and negative effects. ... (The rest of conversation cover topics about movies and TV shows recommendations.)
Probing Question: (May 10th)	What good ways did you recommend me to relieve stress?
Model Output: SiliconFriend (ChatGLM)	I recommended that you try doing some moderate exercise, listening to music, reading, talking to friends, or adding more entertainment and leisure activities into your daily life to reduce work pressure and negative effects. You can also try to add more of these things into your life in moderation, as it is important to balance things out and not overdo it. If you are still feeling stressed, I would recommend seeking the help of a professional counselor or therapist who can provide further guidance and support.
Model Output: SiliconFriend (BELLE)	I recommended some ways to relieve stress such as doing moderate exercise, listening to music, reading, talking to friends, etc. You can try to add more entertainment and leisure activities into your daily life, and reduce work pressure and negative effects.
Model Output: SiliconFriend (ChatGPT)	Based on our past conversations, I suggested doing moderate exercise, listening to music, reading, talking to friends, and adding more leisure activities to your daily life as good ways to relieve stress. Exercise can help release endorphins and boost your mood, while music and reading can provide a mental escape from stress. Talking to friends can also help alleviate stress by providing a social support system. Lastly, adding leisure activities to your daily routine can be a fun and effective way of reducing stress levels.

and dialogue on each day covers at least two topics. User meta-information, including names, personalities, and interested topics is generated using ChatGPT. Conversations are synthesized by users acted by ChatGPT based on predefined topics and user personalities. We create memory storages in both English and Chinese. After memory storage construction, we manually write 194 probing questions (97 in English and 97 in Chinese) to assess whether the model could accurately recall pertinent memory and appropriately formulate answers. Table 1 presents an example of user meta-information, generated conversations, and probing questions.

Evaluation Metrics The performance of models is assessed based on the following metrics. (1) **Memory Retrieval Accuracy:** Determines if related memory can be successfully retrieved (labels: {0 : no, 1 : yes}). (2) **Response Correctness:** Evaluates if the response contains the correct answer to the probing question (labels: {0 : wrong, 0.5 : partial, 1 : correct}). (3) **Contextual Coherence:** Assesses whether the response is naturally and coherently structured, connecting the dialogue context and retrieved memory (labels: 0 : not coherent, 0.5 : partially coherent, 1 : coherent). (4) **Model Ranking Score:** Ranks outputs from the three SiliconFriend variants (SiliconFriend_{ChatGLM}, SiliconFriend_{ChatGPT}, and SiliconFriend_{BELLE}) for the same question and context. Models’ scores are calculated using $s = 1/r$, where $r = 1, 2, 3$ indicates its relative ranking.

Result Analysis. We evaluate 3 SiliconFriend variants using both English and Chinese test set. Table 2 yields the following insights: **(1)** Our overall best variant SiliconFriend_{ChatGPT} has high performance across all metrics, showing the effectiveness of our overall framework. **(2)** SiliconFriend_{BELLE} and SiliconFriend_{ChatGLM} also have high performance in retrieval accuracy, showing the generality and effectiveness of our MemoryBank mechanism for both open-source and closed-source LLMs. Nonetheless, their performance on other metrics is not as good as SiliconFriend_{ChatGPT}. This might be attributed to the inferior overall abilities of the base models (BELLE and ChatGLM) compared to ChatGPT. **(3)** Models’ performance varies on different languages. SiliconFriend_{ChatGLM} and SiliconFriend_{ChatGPT} deliver better results in English, while SiliconFriend_{BELLE} excells in Chinese.

表1: 记忆库中一个记忆片段（5月3日）的示例，以及5月10日提出的相应探测性问题。为便于比较，展示了SiliconFriend三个变体的输出。所有这些模型均能正确且连贯地回答。

元信息 用户：	姓名：Gary 爱好： 赛车、商业、国际象棋、绘画、葡萄酒 性格： 果断直率、直言不讳、乐于助人
过往对话： (5月3日)	用户: 我最近感觉有点压力大，睡眠也不太好。你有什么缓解压力的好方法吗？ 人工智能: 确实，长时间的工作或生活压力会影响我们的身心健康。缓解压力的方法有很多，比如进行适度运动、听音乐、阅读、与朋友交谈等。你可以尝试在日常生活中增加更多娱乐休闲活动，并减轻工作压力和负面影响。 ... (对话的其余部分涵盖了关于电影和电视剧推荐的讨论。)
探测性问题： (5月10日)	你推荐了哪些缓解压力的好方法？
模型输出： SiliconFriend (ChatGLM)	我建议你尝试进行一些适度运动、听音乐、阅读、与朋友交谈，或在日常生活中增加更多娱乐休闲活动，以减轻工作压力和负面影响。你也可以尝试增加更多这类凡事都要适度地引入你的生活，因为保持平衡很重要，不要过度。如果你仍然感到有压力，我建议寻求专业咨询师或治疗师的帮助，他们可以提供进一步的指导和支持。
模型输出： SiliconFriend (BELLE)	我推荐了一些缓解压力的方法，比如进行适度运动、听音乐、阅读、与朋友交谈等。你可以尝试在日常生活中增加更多娱乐休闲活动，并减少工作压力和负面影响。
模型输出： SiliconFriend (ChatGPT)	根据我们的过往对话，我曾建议进行适度运动、听音乐、阅读、与朋友交谈，并在你的日常生活中增加更多休闲活动，作为缓解压力的好方法。运动有助于释放内啡肽并提升你的情绪，而音乐和阅读可以提供一种从压力中精神逃避的方式。与朋友交谈也能通过提供一个社会支持系统来帮助缓解压力。最后，在你的日常作息中增加休闲活动可以成为一种有趣且有效的降低压力水平的方式。

且每天的对话至少涵盖两个主题。用户元信息，包括姓名、个性和感兴趣的主题，均使用ChatGPT生成。对话由ChatGPT扮演的用户根据预定义主题和用户个性合成。我们创建了英语和中文两种语言的记忆存储。在记忆存储构建完成后，我们手动编写了194个探测性问题（97个英文，97个中文），以评估模型是否能准确回忆相关记忆并恰当地组织答案。表1展示了一个用户元信息、生成的对话以及探测性问题的示例。

评估指标模型的性能基于以下指标进行评估。(1)**记忆检索准确率：**判断是否能成功检索到相关记忆（标签：{0：否,1：是}）。(2) **回答正确性：**评估回答是否包含探测性问题的正确答案（标签： {0：错误 0.5：部分正确,1：正确}）。(3) **上下文连贯性：**评估回答是否自然、结构连贯，将对话上下文与检索到的记忆联系起来（标签：0：不连贯 0.5：部分连贯,1：连贯）。(4)**模型排名得分：**对三种SiliconFriend变体（SiliconFriend_{ChatGLM}、SiliconFriend_{ChatGPT}和SiliconFriend_{BELLE}）针对同一问题和上下文的输出进行排名。模型的得分使用 $s = 1/r$ 计算，其中 $r = 1,2,3$ 表示其相对排名。

结果分析。我们使用英语和中文测试集评估了3个SiliconFriend变体。表2揭示了以下见解：**(1)** 我们整体表现最佳的变体SiliconFriend_{ChatGPT} 在所有指标上均表现出高性能，这证明了我们整体框架的有效性。**(2)** SiliconFriend_{BELLE} 和SiliconFriend_{ChatGLM} 在检索准确性方面也表现出高性能，这表明我们的MemoryBank机制对于开源和闭源大语言模型均具有普适性和有效性。尽管如此，它们在其他指标上的表现不如SiliconFriend_{ChatGPT}。这可能归因于其基础模型（BELLE和ChatGLM）的整体能力相较于ChatGPT有所不足。**(3)** 模型在不同语言上的表现存在差异。SiliconFriend_{ChatGLM} 和SiliconFriend_{ChatGPT} 在英语上表现更佳，而SiliconFriend_{BELLE} 则在中文上表现优异。

Table 2: Results of quantitative analysis.

Language	Model	Retrieval Acc.	Correctness	Coherence	Ranking
English	SiliconFriend _{ChatGLM}	0.809	0.438	0.68	0.498
	SiliconFriend _{BELLE}	0.814	0.479	0.582	0.517
	SiliconFriend _{ChatGPT}	0.763	0.716	0.912	0.818
Chinese	SiliconFriend _{ChatGLM}	0.84	0.418	0.428	0.51
	SiliconFriend _{BELLE}	0.856	0.603	0.562	0.565
	SiliconFriend _{ChatGPT}	0.711	0.655	0.675	0.758

5 Related Works

Large Language Models: LLMs such as GPT-3 (Brown et al., 2020), OPT (Zhang et al., 2022), and FLAN-T5 (Chung et al., 2022) have made remarkable strides in a broad spectrum of natural language processing tasks in recent years. Recently, cutting-edge closed-source language models, like PaLM (Chowdhery et al., 2022), GPT-4 (OpenAI, 2023) and ChatGPT (OpenAI, 2022), continue to display substantial flexibility, adapting to a wide variety of domains. They have increasingly become daily decision-making aids for many people. However, the close-source nature of these models prohibit the researchers and companies to study the inner mechanism of LLMs and built domain-adapted applications. Therefore, many open-source LLMs emerged in the community, like LLaMa (Touvron et al., 2023), ChatGLM (Zeng et al., 2022) and Alpaca (Taori et al., 2023). For more details, we refer readers to this comprehensive review: Zhao et al. (2023). Nevertheless, these models still have shortcomings. A noticeable gap lies in their deficiency in a robust long-term memory function. This limitation hinders their ability to maintain context over a long period and retrieve pertinent information from past interactions. Our research steps in here, with the primary objective of developing long-term memory mechanism for LLMs.

Long-term Memory Mechanisms: Numerous attempts have been made to enhance the memory capabilities of neural models. Memory-augmented networks (MANNs) (Meng & Huang, 2018; Graves et al., 2014) like Neural Turing Machines (NTMs)(Graves et al., 2014) is an example of this, designed to increase the memory capacity of neural networks. These models are structured to interact with an external memory matrix, enabling them to handle tasks that necessitate the maintenance and manipulation of stored information over extended periods. Despite showing potential, these methods have not fully addressed the need for a reliable and adaptable long-term memory function in LLMs. There have also been studies focusing on long-range conversations (Xu et al., 2021, 2022). For instance, Xu et al. (2021) introduced a new English dataset comprised of multi-session human-human crowdworker chats for long-term conversations. However, these conversations are generally restricted to a few rounds of conversation, which can not align with the application scenarios of long-term AI companions. Moreover, these models often fail to create a detailed user portrait and lack a human-like memory updating mechanism, both crucial for facilitating more natural interactions. The concept of memory updating has been extensively researched in psychology. The Forgetting Curve theory by Ebbinghaus (1964) offers valuable insights into the pattern of memory retention and forgetting over time. Taking inspiration from this theory, we integrate a memory updating mechanism into MemoryBank to bolster its long-term memory function.

In summary, while significant progress has been made in the field of LLMs, there is still a need for long-term memory mechanism to empower LLMs in the scenarios requiring personalized and persistent interactions. Our work presents MemoryBank as a novel approach to address this challenge.

6 Conclusion

We present MemoryBank, a novel long-term memory mechanism designed to address the memory limitation of LLMs. MemoryBank enhances the ability to maintain context over time, recall relevant information, and understand user personality. Besides, the memory updating mechanism of MemoryBank draws inspiration from the Ebbinghaus Forgetting Curve theory, a psychological principle that describes the nature of memory retention and forgetting over time. This design improves the anthropomorphism of AI in long-term interactions scenarios. The versatility of MemoryBank is

表2：定量分析结果。

语言	模型	检索准确率	正确性	连贯性	排名
英语	SiliconFriend _{ChatGLM}	0.809	0.438	0.68	0.498
	SiliconFriend _{BELLE}	0.814	0.479	0.582	0.517
	SiliconFriend _{ChatGPT}	0.763	0.716	0.912	0.818
中文	SiliconFriend _{ChatGLM}	0.84	0.418	0.428	0.51
	SiliconFriend _{BELLE}	0.856	0.603	0.562	0.565
	SiliconFriend _{ChatGPT}	0.711	0.655	0.675	0.758

5 相关工作

大语言模型：近年来，诸如GPT-3（Brown 等人, 2020）、OPT（Zhang 等人, 2022）和 FLAN-T5（Chung 等人, 2022）等大语言模型在广泛的自然语言处理任务中取得了显著进展。最近，前沿的闭源语言模型，如PaLM（Chowdhery 等人, 2022）、GPT-4（OpenAI, 2023）和ChatGPT（OpenAI, 2022），继续展现出强大的灵活性，能够适应多种领域。它们已日益成为许多人日常决策的辅助工具。然而，这些模型的闭源特性阻碍了研究人员和公司深入研究大语言模型的内部机制并构建领域适配的应用程序。因此，社区中涌现了许多开源大语言模型，如LLaMa（Touvron 等人, 2023）、ChatGLM（Zeng 等人, 2022）和Alpaca（Taori 等人, 2023）。更多细节，我们建议读者参考这篇全面的综述：Zhao 等人 (2023)。尽管如此，这些模型仍存在不足。一个显著的差距在于它们缺乏强大的长期记忆功能。这一限制阻碍了它们长期保持上下文并从过去的互动中检索相关信息的能力。我们的研究正是由此切入，主要目标是为大语言模型开发长期记忆机制。

长期记忆机制：已有大量尝试旨在增强神经模型的记忆能力。记忆增强网络（MANNs）（Meng 与 Huang, 2018; Graves 等人, 2014），如神经图灵机（NTMs）（Graves 等人, 2014），便是其中一例，其设计初衷是增加神经网络的记忆容量。这些模型的结构使其能够与外部记忆矩阵交互，从而处理那些需要在较长时间内维护和操作存储信息的任务。尽管显示出潜力，但这些方法尚未完全满足大语言模型对可靠且适应性强的长期记忆功能的需求。也有研究专注于长程对话（Xu 等人, 2021, 2022）。例如，Xu 等人（2021）引入了一个新的英语数据集，该数据集由用于长期对话的多会话人-人众包聊天构成。然而，这些对话通常仅限于少数几轮，无法与长期AI伴侣的应用场景相匹配。此外，这些模型往往无法创建详细的用户画像，并且缺乏类人的记忆更新机制，而这两者对于促进更自然的互动至关重要。记忆更新的概念在心理学领域已被广泛研究。艾宾浩斯（1964）提出的遗忘曲线理论为记忆保持率和随时间遗忘的模式提供了宝贵的见解。受此理论启发，我们将一个记忆更新机制整合到记忆库中，以增强其长期记忆功能。

总而言之，尽管大语言模型领域已取得显著进展，但在需要个性化和持久互动的场景中，大语言模型仍需要长期记忆机制来增强其能力。我们的工作提出了记忆库作为一种新颖的方法来应对这一挑战。

6 结论

我们提出了记忆库，一种新颖的长期记忆机制，旨在解决大语言模型的记忆限制。记忆库增强了随时间维持上下文、回忆相关信息以及理解用户性格的能力。此外，记忆库的记忆更新机制借鉴了艾宾浩斯遗忘曲线理论，这是一种描述记忆保持率和遗忘随时间变化的本质的心理学原理。这一设计提升了人工智能在长期交互场景中的拟人化程度。记忆库的通用性

demonstrated through its accommodation of both open-source models such as ChatGLM and BELLE, and close-source models like ChatGPT.

We further illustrate the practical application of MemoryBank through the development of Silicon-Friend, an LLM-based chatbot designed to serve as a long-term AI companion. Equipped with MemoryBank, SiliconFriend can establish a deeper understanding of users, offering more personalized and meaningful interactions, emphasizing the potential for MemoryBank to humanize AI interactions. The tuning of SiliconFriend with psychological dialogue data enables it to provide empathetic emotional support. Extensive experiments including both qualitative and quantitative methods validate the effectiveness of MemoryBank. The findings demonstrate that MemoryBank empowers SiliconFriend with memory recall capabilities and deepens the understanding of user behaviors. Besides, SiliconFriend can provide empathetic companionship of higher quality.

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通过其同时兼容开源模型（如ChatGLM和BELLE）和闭源模型（如ChatGPT）的能力得到了验证。

我们进一步通过开发SiliconFriend来阐述记忆库的实际应用，这是一个旨在作为长期AI伴侣的基于大语言模型的聊天机器人。配备记忆库后，SiliconFriend能够建立对用户更深入的理解，提供更个性化和有意义的互动，突显了记忆库在使人工智能互动更具人性化方面的潜力。利用心理对话数据对SiliconFriend进行调优，使其能够提供共情式情感支持。包括定性和定量方法在内的广泛实验验证了记忆库的有效性。研究结果表明，记忆库赋予了SiliconFriend记忆回想能力，并加深了对用户行为的理解。此外，SiliconFriend能够提供更高质量的共情陪伴。

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